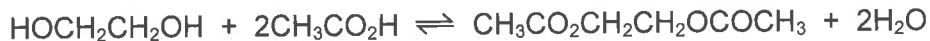


- 25** When a mixture of 1 mol of ethane-1,2-diol and 2 mol of ethanoic acid is allowed to reach equilibrium at a given temperature, the mixture is found to contain  $x$  mol of the diol.



You should assume that these are the only four species present at equilibrium.

What are the amounts of the four species at equilibrium?

|          | diol / mol | acid / mol | ester / mol | water / mol |
|----------|------------|------------|-------------|-------------|
| <b>A</b> | $x$        | $2x$       | $1 - x$     | $2(1 - x)$  |
| <b>B</b> | $x$        | $2 - x$    | $1 - x$     | $2 - x$     |
| <b>C</b> | $x$        | $2x$       | $x$         | $2x$        |
| <b>D</b> | $x$        | $2(1 - x)$ | $x$         | $2x$        |